

1210. Minimum Moves to Reach Target with Rotations

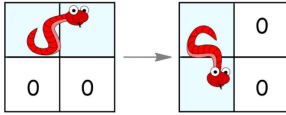
Solved ●

Hard Topics Hint

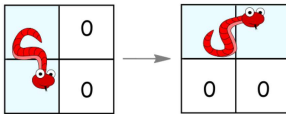
In an $n \times n$ grid, there is a snake that spans 2 cells and starts moving from the top left corner at $(0, 0)$ and $(0, 1)$. The grid has empty cells represented by zeros and blocked cells represented by ones. The snake wants to reach the lower right corner at $(n-1, n-2)$ and $(n-1, n-1)$.

In one move the snake can:

- Move one cell to the right if there are no blocked cells there. This move keeps the horizontal/vertical position of the snake as it is.
- Move down one cell if there are no blocked cells there. This move keeps the horizontal/vertical position of the snake as it is.
- Rotate clockwise if it's in a horizontal position and the two cells under it are both empty. In that case the snake moves from (r, c) and $(r, c+1)$ to (r, c) and $(r+1, c)$.



- Rotate counterclockwise if it's in a vertical position and the two cells to its right are both empty. In that case the snake moves from (r, c) and $(r+1, c)$ to (r, c) and $(r, c+1)$.



Return the minimum number of moves to reach the target.

If there is no way to reach the target, return -1 .

Example 1:

START					
		0	0	0	1
1	1	0	0	1	0
0	0	0	0	1	1
0	0	1	0	1	0
0	1	1	0	0	0
0	1	1	0	0	0
					FINISH

Input: grid = $[[0,0,0,0,0,1],$
 $[1,1,0,0,1,0],$
 $[0,0,0,0,1,1],$
 $[0,0,1,0,1,0],$
 $[0,1,1,0,0,0],$
 $[0,1,1,0,0,0]]$

Output: 11

Explanation:

One possible solution is [right, right, rotate clockwise, right, down, down, down, down, rotate counterclockwise, right, down].

Example 2:

Input: grid = $[[0,0,1,1,1,1],$
 $[0,0,0,0,1,1],$
 $[1,1,0,0,0,1],$
 $[1,1,1,0,0,1],$
 $[1,1,1,0,0,1],$
 $[1,1,1,0,0,0]]$

Output: 9

Constraints:

- `2 <= n <= 100`
- `0 <= grid[i][j] <= 1`
- It is guaranteed that the snake starts at empty cells.

Seen this question in a real interview before? 1/6

Yes No

Accepted 13,760/25.9K | Acceptance Rate 53.0%

 Topics	▼
	▼
 Hint 1	▼
 Hint 2	▼
 Discussion (20)	▼